

NPTEL Course on
"Computer System Optimization for ML"

Lecture #44

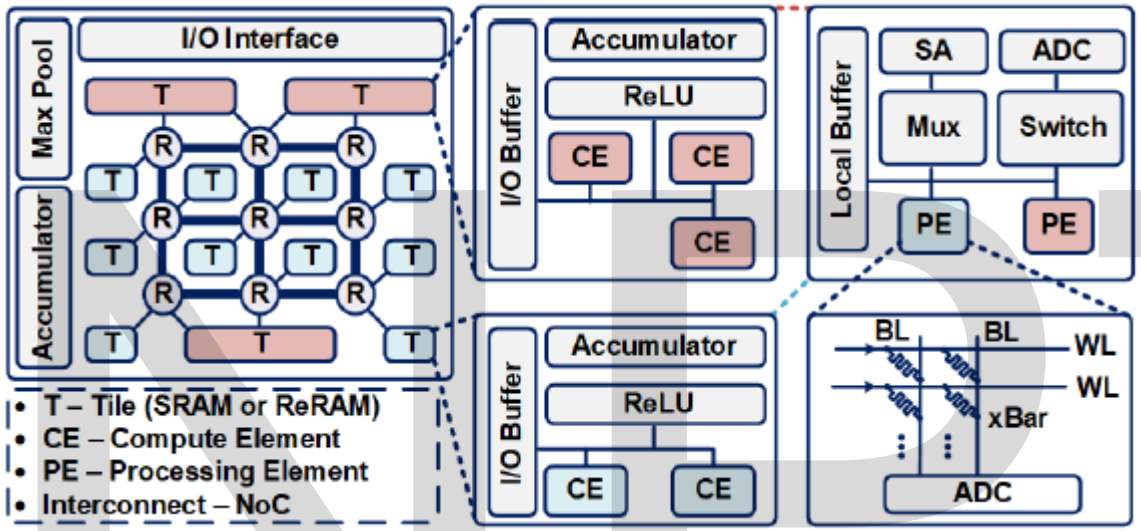
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Agenda

- Recap
- Communication-aware Optimization
- Achieving minimum communication latency

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Reducing Residual Area



Symbol	Definition	Symbol	Definition
T_k	Number of tiles in k^{th} layer	N_{bits}	Weight precision
Kx_k, Ky_k	Kernel size	c_k	Number of CEs in T_k
$(PE_x)_k$, $(PE_y)_k$	Size of the PE array	p_k	Number of PEs in c_k
I_k	Input activations in k^{th} layer	N_k^{if}, N_k^{of}	Number of i/p and o/p features

$$N_k^r = \left\lceil \frac{Kx_k \times Ky_k \times N_k^{if}}{(PE_x)_k} \right\rceil$$ → Required number of rows of PEs

$$N_k^c = \left\lceil \frac{N_k^{of} \times N_{bits}}{(PE_y)_k} \right\rceil$$ → Required number of columns of PEs

$$T_k = \left\lceil \frac{N_k^r \times N_k^c}{c_k \times p_k} \right\rceil$$

$$(c_k \times p_k \times T_k - N_k^r \times N_k^c)$$ → Residual Area

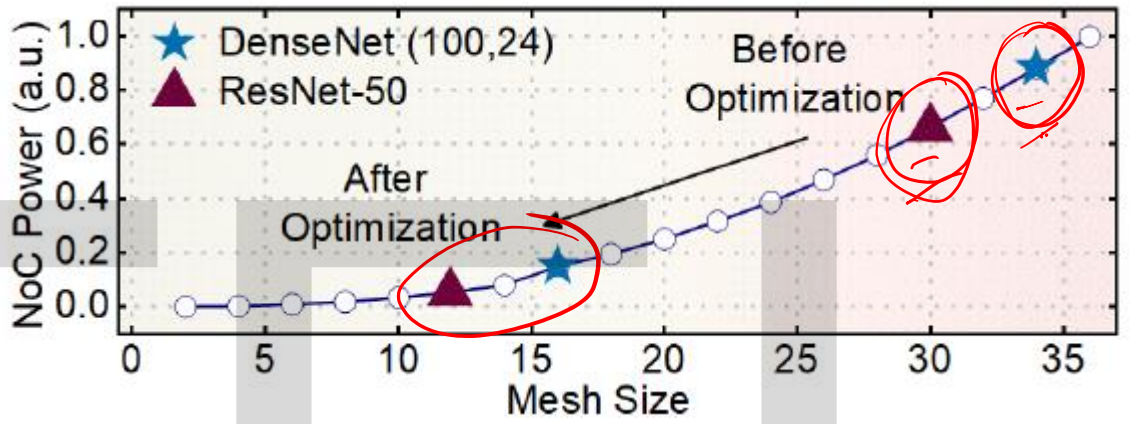
$$A_k(c_k, p_k) = (c_k \times p_k \times T_k - N_k^r \times N_k^c) \times T_k^2$$ → Cost function

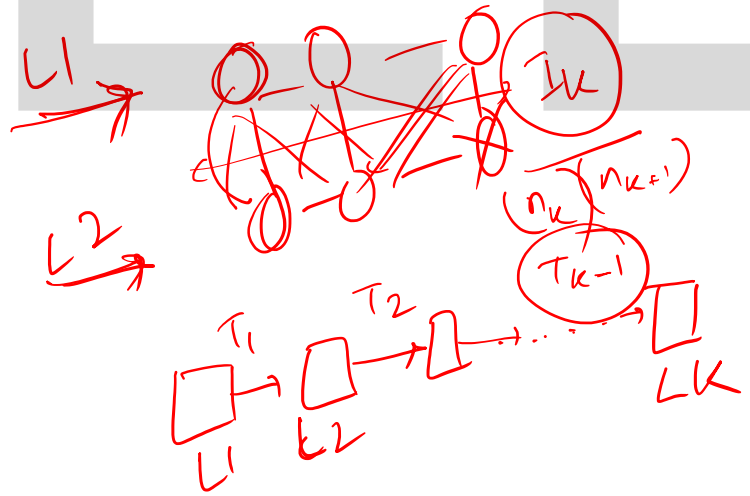
$$\underset{c_k, p_k}{\text{minimize}} \quad A_k(c_k, p_k);\ k = 1, \ldots, K,$$

$$\text{subject to} \begin{cases} c_{min} \le c_k \le c_{max}, \\ p_{min} \le p_k \le p_{max}. \end{cases}$$ → Constrained optimization

Area-aware Optimization is NOT Sufficient (1)

- NoC Size is too big!
  - If each tile is connected to a router
- Objective function to represent NoC energy
  - n_k: number of NoC routers for layer-k
  - I_k: number of activations between layer-k and layer-(k+1)
  - Number of activations between each src-dest pair of router
    - $\frac{I_k}{n_k n_{k+1}}$
  - Total communication volume
    - $\sum_{k=1}^{K-1} \frac{I_k}{n_k n_{k+1}}$





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Area-aware Optimization is NOT Sufficient (2)

- **Objective function**

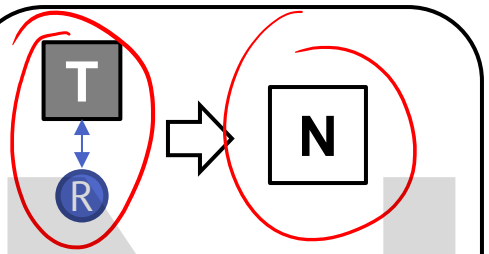
$$-E(\bar{n}) = \sum_{k=1}^{K-1} \left(\frac{I_k}{n_k n_{k+1}} \right) \left(\sum_{k=1}^{K-1} n_k \right)$$

- **Constrained optimization**

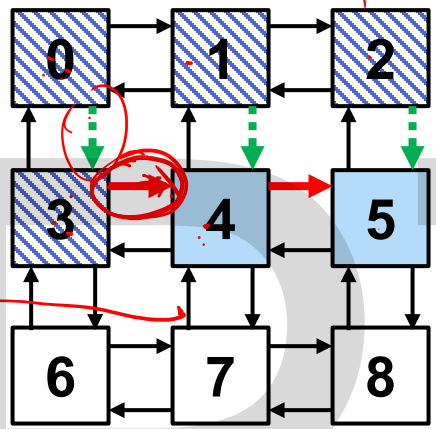
$$\begin{aligned} \text{minimize} \quad & E(\bar{\mathbf{n}}) \\ \bar{\mathbf{n}} \\ \text{subject to} \quad & n_k \geq 1; \quad k = 1, \ldots, K, \\ & \sum_{k=1}^{K} n_k \leq N. \end{aligned}$$

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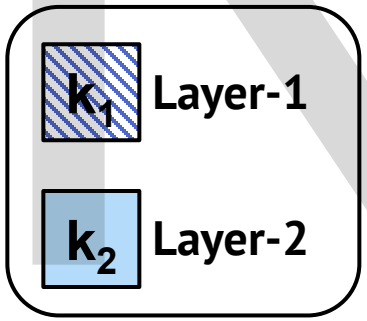
Congestion-free Communication for DNN (1)



One node denotes combination of one IMC tile and a router



- Colored arrows show the link used for the communication
- Dashed (green) arrows show the links with no congestion
- Solid (red) arrows show the links with congestions
  - Example: The link between Node 3 and Node 4 is used by 0-4 and 3-4 traffic



Assumptions

- Y-X routing for NoC
- Each tiles generates P packets to each other tile
- Layer-1 is mapped into Node 0-3; Layer-2 is mapped into Node 4-5
- Communication between Layer-1 and Layer-2

Schedule	Src→Dest
Cycle-1--Cycle-P	0→3
Cycle-1--Cycle-P	3→4
Cycle-1--Cycle-P	1→5
Cycle-1--Cycle-P	2→5
(P + 1)--(2P)	0→4
(P + 1)--(2P)	4→5

Should not 'overlap'!!

Congestion-free Communication for DNN (2)

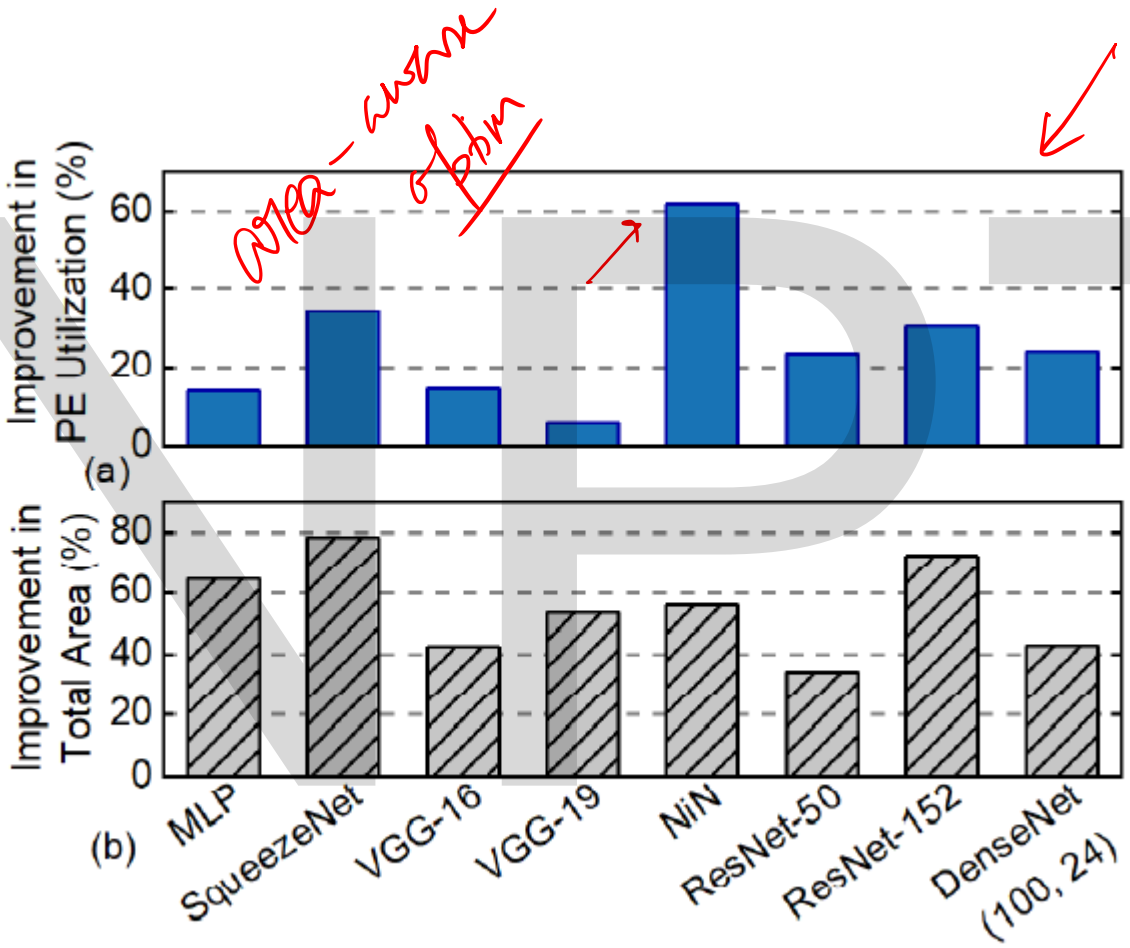
- **Packet scheduling in NoC: minimizing latency**
  - **Making sure there is no congestion in the NoC**
  - **Models each src-dest transaction as separate task**
- **Constructed the constraints**
- **Solved through IBM Cplex to minimize finish time of all tasks**
  - **To obtain start time of each task**

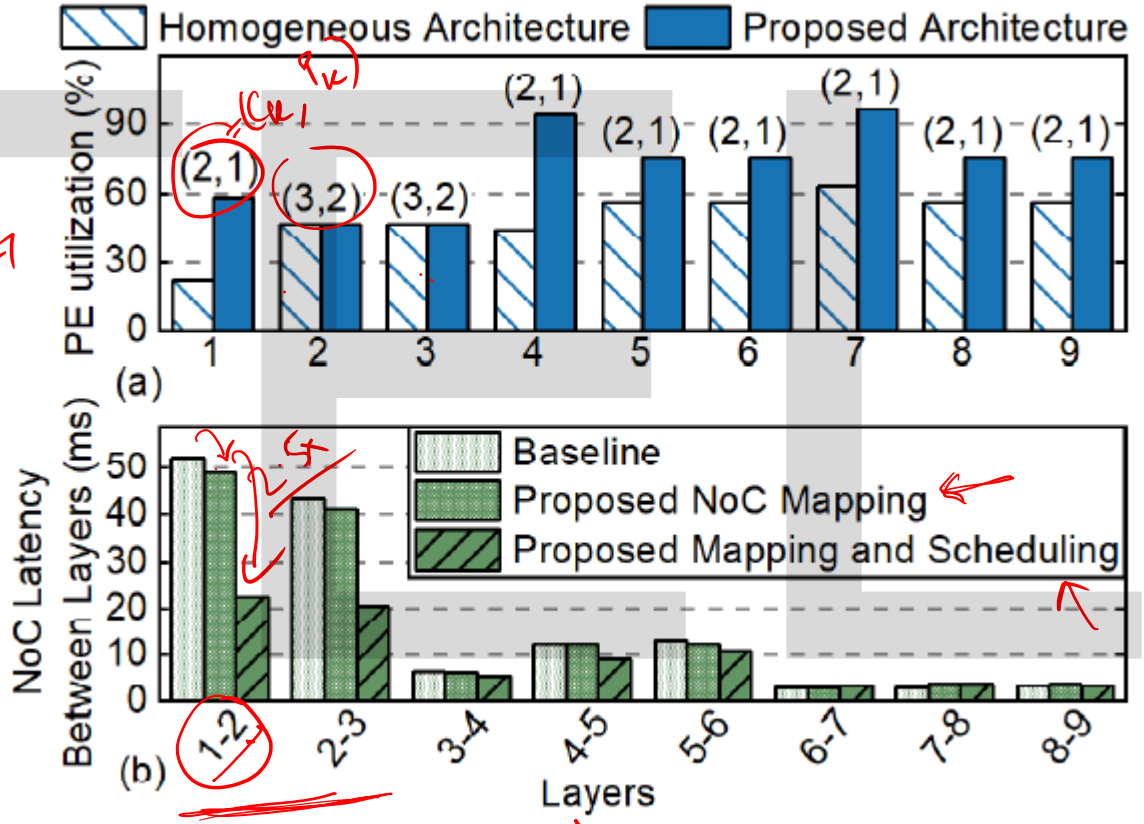
Symbol	Meaning
$t_{terminal}$	Total communication time
C	Set of all non overlapping tasks
τ	Set of all tasks
t_{sd}	The task corresponding to the transaction from source s to destination d

$$\begin{aligned}
\text{minimize} \quad & t_{terminal} \\
\text{subject to} \quad & t_{mn} > t_{pq} + n_{pq} \vee t_{pq} > t_{mn} + n_{mn} \\
& \forall t_{mn}, t_{pq} \in \mathcal{C}, \\
& t_{xy} \geq 0, \forall\ t_{xy} \in \mathcal{T} \\
& t_{terminal} > t_{xy} + n_{xy}, \forall\ t_{xy} \in \mathcal{T}.
\end{aligned}$$

- t_{mn} and t_{pq} must be separated by corresponding number of packets
- One can start only if other one finishes

Results


Improvement in PE Utilization (%)
60
40
20
0
(a)
Improvement in Total Area (%)
80
60
40
20
0
(b)
MLP
SqueezeNet
VGG-16
VGG-19
NiN
ResNet-50
ResNet-152
DenseNet (100, 24)


Homogeneous Architecture
Proposed Architecture
PE utilization (%)
90
60
30
0
(2,1)
(3,2)
(3,2)
(2,1)
(2,1)
(2,1)
(2,1)
(2,1)
(2,1)
1
2
3
4
5
6
7
8
9
(a)
NoC Latency Between Layers (ms)
50
40
30
20
10
0
Baseline
Proposed NoC Mapping
Proposed Mapping and Scheduling
1-2
2-3
3-4
4-5
5-6
6-7
7-8
8-9
(b)
Layers

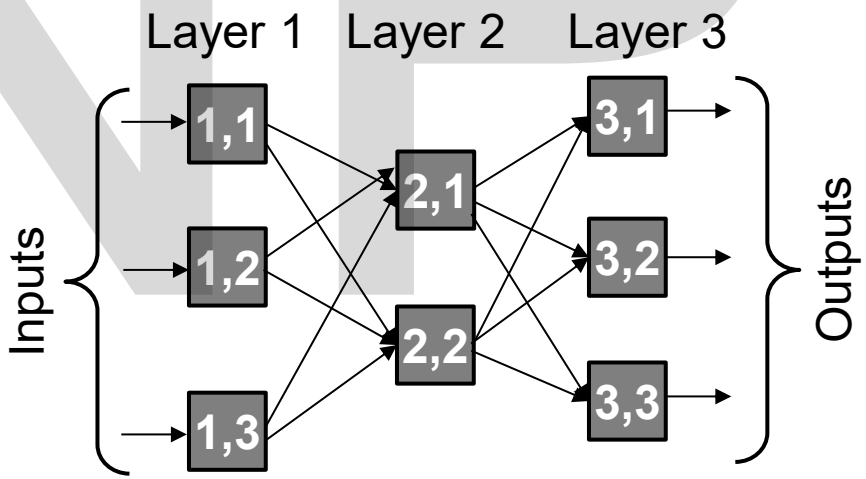
Agenda

- Recap
- Communication-aware Optimization
- **Achieving minimum communication latency**

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Achieving Minimum Possible Communication Latency: Example

- **Target: to achieve minimum possible communication latency for a given DNN on IMC-based hardware platform**
- **Example of *minimum possible communication latency***



(a)

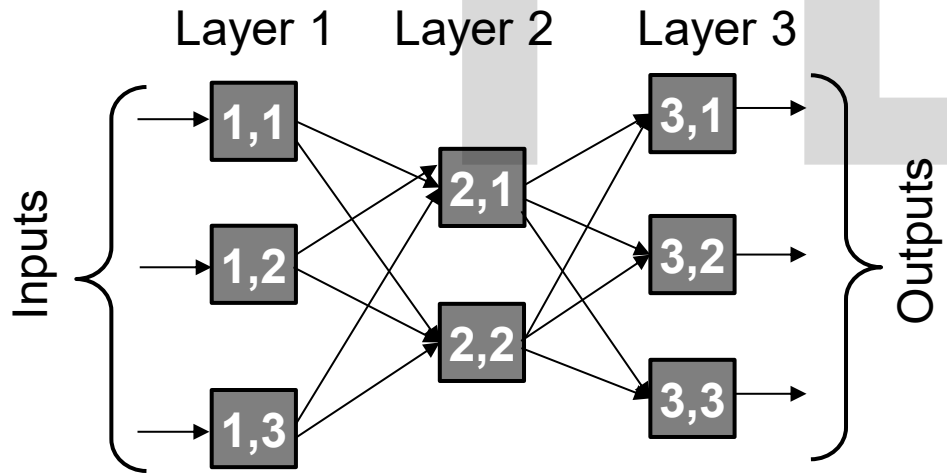
Layer 1 – Layer 2		Layer 2 – Layer 3	
Link	Cycle	Link	Cycle
$L_{1,1} - L_{2,1}$	1	$L_{2,1} - L_{3,1}$	4
$L_{1,2} - L_{2,2}$	1	$L_{2,2} - L_{3,2}$	4
$L_{1,1} - L_{2,2}$	2	$L_{2,1} - L_{3,2}$	5
$L_{1,3} - L_{2,1}$	2	$L_{2,2} - L_{3,1}$	5
$L_{1,2} - L_{2,1}$	3	$L_{2,2} - L_{3,2}$	6
$L_{1,3} - L_{2,2}$	3	$L_{2,1} - L_{3,3}$	6

(b)

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Key Insights

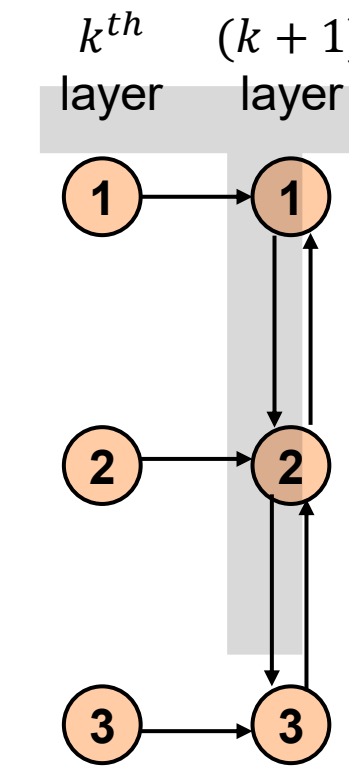
- A single router can not send/receive more than one packet in a cycle. However, multiple routers can send/receive packets in parallel.
- Since a router can send/receive only one packet in a cycle, the congestion is minimum i.e., no more than a transaction is scheduled through a particular link.



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Solution: Custom Interconnect (A Simple Case)

- Two consecutive layers consists of equal number of routers
  - Three routers in this case
- Each router can send/receive only one packet at one cycle
- In one round of communication each routers in k^{th} layer send a packet to each router in $(k+1)^{th}$ layer
- One transaction is finished in 3 cycles
  - Minimum possible latency



(a)

Link	Cycle	Transaction
$L_{k,1} - L_{(k+1),1}$	1	$R_{k,1} - R_{(k+1),1}$
$L_{k,2} - L_{(k+1),2}$	1	$R_{k,2} - R_{(k+1),2}$
$L_{k,3} - L_{(k+1),3}$	1	$R_{k,3} - R_{(k+1),3}$
$L_{(k+1),1} - L_{(k+1),2}$	2	$R_{k,1} - R_{(k+1),2}$
$L_{(k+1),2} - L_{(k+1),3}$	2	$R_{k,2} - R_{(k+1),3}$
$L_{(k+1),3} - L_{(k+1),2}$	2	$R_{k,3} - R_{(k+1),2}$
$L_{(k+1),2} - L_{(k+1),1}$	2	$R_{k,2} - R_{(k+1),1}$
$L_{(k+1),2} - L_{(k+1),3}$	3	$R_{k,1} - R_{(k+1),3}$
$L_{(k+1),2} - L_{(k+1),1}$	3	$R_{k,3} - R_{(k+1),1}$

(b)

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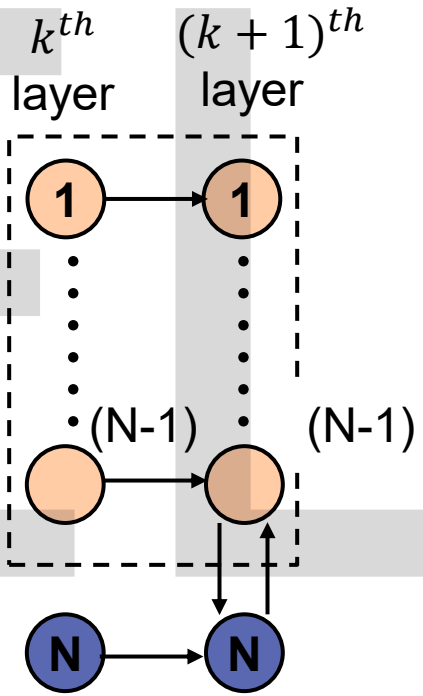
Proof by Induction

New Transactions:

Horizontal Link (L_{1N}-L_{2N}) will be active on cycle 1 which does not overlap with any of the transaction of the configuration with (N-1) routers

Upward Vertical Link: New transactions will carry the packet from the router at the position (1,N). The vertical link ($L_{2,N}$-$L_{2,(N-1)}$) will be activated on cycle 1. All other new transactions will happen 1 cycle after the last transaction for the configuration with (N-1) routers.

Downward Vertical Link: New transactions will happen only in ($L_{2,(N-1)} - L_{2,N}$)



Schedules in general:

Horizontal Link: $L_{k,n}$-$L_{k+1,n}$ in cycle 1 carries transaction $T_{k,n}$-$T_{k+1,n}$

Upward Vertical Link: $L_{k+1,n}$-$L_{k+1,n-1}$ in m^{th} cycle carries the transaction

$T_{k,n+m-2}$-$T_{k+1,n-1}$, if the transaction exists

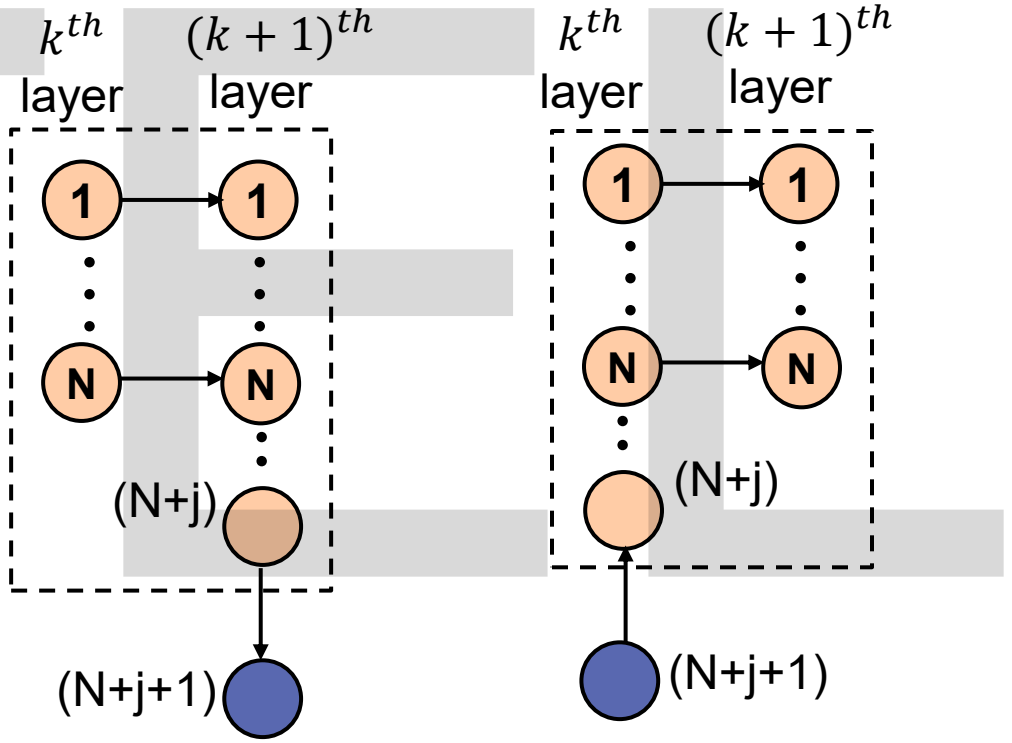
Downward Vertical Link: $L_{k+1,n}$-$L_{k+1,n+1}$ in m^{th} cycle carries the transaction

$T_{k,\,n-m+2}$-$T_{k+1,n+1}$, if the transaction exists

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Other Cases

- Also constructed the schedules when the number of the routers in two consecutive layers are not equal
- Detailed proof is shown in [1]
- We obtain minimum possible communication latency for a give DNN



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Reconfigurable NoC (1)

- **There are two class of CNNs**
  - One is used in edge computing hardware
  - Another is used in cloud-based computing platform
- **We observe similarity in optimal NoC for different networks in each of the two class**
  - Similarity in number of routers for each layer (shown in next slide)
- **On the chip, a number of routers will be reserved for each layer**
- **Before executing the NN algorithm, the proposed algorithm is run (on the chip) to reconfigure the NoC**
  - Deciding the number of routers needed for each layer
  - Deciding the optimal schedule

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Reconfigurable NoC (2)

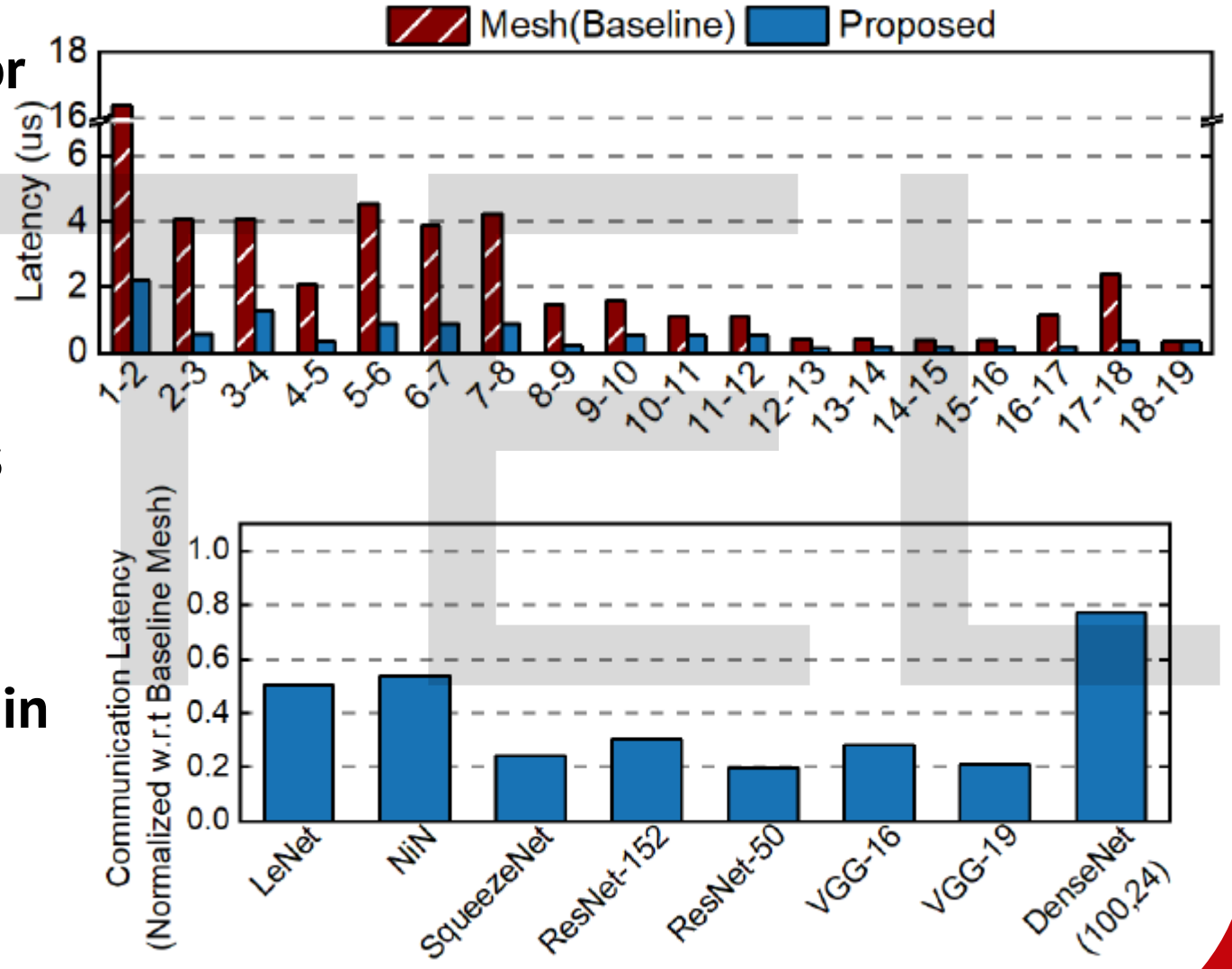
	lenet-5	nin-12	squeeze-26	vgg-16	vgg-19	generic_noc
Layer 1	4	2	5	4	6	6
2	4	2	5	4	6	6
3	4	2	4	4	5	5
4	2	2	6	4	5	6
5		2	6	4	4	6
6		2	4	4	4	4
7		2	4	4	4	4
8		2	4	4	4	4
9		2	3	4	3	4
10		2	4	4	3	4
11			4	4	3	4
12			3	4	3	4
13			3	4	2	4
14			3	2	2	3
15			2	2	2	2
16			3	3	2	3
17			3		2	3
18			2		2	2
19			3		2	3
20			3			3
21			2			2
22			3			3
23			3			3
24			2			2
25			3			3
26			3			3

- **Entries in the table shows optimal number of routers required for different networks (proposed approach)**
- **The right-most column (generic_noc) shows number of routers allocated per layer on the chip**
  - Maximum of all networks for that layer
  - The chip can afford all the 5 CNNs shown in the table
- **At runtime, for a given CNN, number of routers required for each layers of the CNN and the corresponding schedules is computed with the proposed technique**
  - Reconfigurability
- **A given network can use only the maximum available routers for the corresponding layer**
  - Can lead to sub-optimality
  - Should still be optimal than mesh-based solution
- **This hardware can support CNNs with up to 26 layers**
- **Similar common features for cloud-computing based NNs**

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Experimental Results [1]

- **Layer-wise improvements shown for VGG-19**
  - **Up to 8× improvement in communication latency**
- **Improvement in communication latency is shown for different DNNs**
  - **Normalized with respect to the baseline communication latency**
- **Results in up to 25% improvement in inference latency**



[1] Mandal, Sumit K., et al. "A latency-optimized reconfigurable NoC for in-memory acceleration of DNNs." *IEEE Journal on Emerging and Selected Topics in Circuits and Systems* 10.3 (2020): 362-375.

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THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)